OFER SHAHAL

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SUMMARY

Data and geospatial analytics professional with experience in transportation, mobility, and national-scale data projects. Hands-on in Python, SQL, statistical analysis, GIS, and machine learning workflows, with experience working on GTFS, SIRI, micromobility, and spatial-temporal datasets. Combines analytical execution with project and stakeholder leadership, including delivery of national-scale geospatial data systems and data-driven decision-support work in government environments.

SKILLS

Programming: Python, SQL

Data Science & AI: Data Modelling, Pandas, NumPy, ML Regression & Classifiers Models, scikit-learn, Random Forest, XGBoost, SHAP, Optuna, LLM Agents, Tool-Calling, ReAct Agents, Claude

Databases & Data: PostgreSQL, PostGIS, DuckDB, Excel

Geospatial Analytical Tools: QGIS, ArcGIS, GeoPandas, OSMnx, NetworkX, GDAL, Network Analysis, Space Syntax

Transportation: GTFS, SIRI, Micromobility (MDS format), Google API, Routing & Shortest-Path Analysis

Visualization & Mapping: Matplotlib, Seaborn, Streamlit, Plotly, Folium, Leaflet

Professional: Technical Team Leadership, Project Management, Stakeholder Management, Detail-Oriented, Strategic Thinking, Team Player, Self-Learner

EXPERIENCE

Data Systems & Geospatial Project Lead 2025 - Present
Sivan Design D.S LTD, Israel / Zanzibar / Nigeria

- Translating government and operational requirements into data structures, analytical workflows, and system requirements for national geospatial data programs.
- Leading cross-functional technical delivery across developers, GIS specialists, and client stakeholders, aligning data architecture and platform capabilities with business and operational needs.
- Designing data schemas, metadata structures, and system architecture for national-scale spatial data infrastructure (SDI) programs (ZanSDI, Ogun State SDI), including deployment of an open-source geospatial stack (GeoNode, GeoServer, PostGIS).
- Preparing technical documentation, including data model design and integration specifications for multi-machine environments.

Transportation & Geospatial Data Analyst 2022 - 2025
Tel Aviv-Yafo Municipality, Tel Aviv-Yafo

- Automated data processes integrating GTFS, SIRI, and e-scooter datasets to support transportation analytics.
- Applied statistical and spatial analysis to uncover mobility patterns, delivering dashboards that informed planning decisions.

GIS Academic Teaching Assistant 2024
Tel Aviv University, Tel-Aviv
Preparing Basic and Advanced GIS practical courses in QGIS and ArcGIS, and delivering them.

Geospatial Analyst & Researcher 2023
Freelance
Conducted freelance research commissioned by the Tel Aviv-Yafo Municipality, analyzing ~1M shared e-scooter trips using statistical modeling and spatial analysis to inform micro-mobility parking policy.

Research Assitant Geo-Simulation Lab 2021 - 2022
Tel Aviv University, Tel-Aviv
Participated in a joint research with the University of Manchester.

Geospatial Data Analyst 2021 - 2022
Competition Authority, Jerusalem

- Applied geographic principles in departmental research projects.
- Utilized GIS tools and methodologies, to enhance desicion-making.

GIS Operator 2020 - 2021
Ministry of Environmental Protection, Jerusalem
Managed and enhanced GIS data for the Environmental Planning Dept.

EDUCATION

Certified Data Scientist with Expertise in Generative AI and AI Agents 2025 - 2026
Bar-Ilan University

Master's Degree in Geography 2022 - 2024
Tel-Aviv University
Thesis in Geosimulation lab.

Bachelor's Degree in Geography & Middle Eastern Studies 2018 - 2022
Tel-Aviv University

PROJECTS

Predicting Bus Trip Performance with ML
Built an XGBoost classifier to predict bus delays using GTFS, SIRI, and weather features; used Optuna for tuning and SHAP for feature interpretation (XGBoost, SHAP, Optuna, scikit-learn). [Bar-Ilan Data Science Course, 2026]

Israel Transit AI Agent
Built a ReAct-style LLM agent that answers transit questions over GTFS/SIRI data using tool-calling, DuckDB, and Streamlit (ReAct Agents, LLM Agents, Tool-Calling, DuckDB, Streamlit). [Bar-Ilan Data Science Course, 2026]

Israel Train Stations Dashboard
Interactive dashboard exploring national rail passenger patterns (Pandas, Streamlit, Plotly, Folium). [Bar-Ilan Data Science Course, 2025]

Master's Thesis: Cycling Infrastructure & E-Scooter Behavior
Geosimulation Lab, Tel Aviv University — under the mentorship of Prof. Itzhak Benenson
Statistical analysis of ~1M shared e-scooter trips (Python, GeoPandas, NetworkX, OSMnx, Shortest-Path Analysis, Statistical Modeling). [Master's Thesis, Tel Aviv University, 2023 - 2025]

Tel Aviv Night Parking Demand Modeling
Predictive modeling for night parking demand using historical spatial-temporal datasets (GeoPandas, Shapely, NetworkX, OSMnx). [Bar-Ilan Data Science Course, 2024 - 2025]

AWARDS & HONORS

M.A. Excellence Scholarship, The Exact Science Faculty 2024
Tel Aviv University

B.A. Excellence Track Scholarship 2019 - 2022
Tel Aviv University

The Exact Science Faculty Dean's List 2019 - 2020
Tel Aviv University

Best Papers Section 2023
18th Conference on Location Based Services, Ghent, Belgium
Conference presentation “Shared Dockless E-Scooters in the City: System Analysis and the Novel Management Policy” selected for the Best Papers Section.